

The University of Nottingham Ningbo China

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2020-2021

MACHINE LEARNING

Time allowed: 2 hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer All Questions

Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

Principal Component Analysis (PCA) is one of the most commonly used dimensionality reduction techniques. It aims to explain or summarize the underlying variance–covariance structure of a large set of variables using a small number of linear combinations, called principal components. PCA is an unsupervised learning method that maximizes variance without using class labels.

PCA is an unsupervised learning technique because it does not use class labels and only relies on the variance–covariance structure of the data.

Question 1: [20 marks]

(a) What is PCA? Briefly describe the goals of PCA? Is PCA a supervised learning technique or an unsupervised learning technique? Compare the similarity and difference between feature selection and PCA.

Feature selection reduces dimensionality by selecting a subset of the original features, whereas PCA reduces dimensionality by transforming the original features into a new set of orthogonal variables through linear combinations. Feature selection preserves interpretability, while PCA is generally more effective at removing redundancy.

(b) Referring to decision tree, briefly explain: How to pick the root node? How to use entropy to associate information with probability?

The goal is to have the resulting decision tree as small as possible, in order to improve generalisation and avoid overfitting, following Occam’s Razor principle.

Entropy measures the impurity of a sample of training examples. Entropy uses class probabilities to quantify the uncertainty of a node. When class probabilities are evenly distributed, entropy is high; when one class dominates, entropy is low.

Question 2: [22 marks]

(a) We will use a single Perceptron classifier to classify 2D real number input into two classes (+1 and -1). The training data are listed as follows.

Input	Output
(2,2)	1
(4,-2)	-1
(-6,-2)	-1
(-6,2)	1

Train the Perceptron with online mode using the training data above from the first row to the last row, with the initialization of weights $w=[1\ 0\ 0]^T$ and $\eta=0.5$. For each iteration, calculate the current output of Perceptron (o) and the updated weights (w). Note that you need to show intermediate steps of your calculation. An answer without the intermediate steps will not receive any marks.

When the problem is not linearly separable, perceptron will fail to converge. ADLINE can overcome this difficulty by finding a best fit approximation to the target. Both Perceptron and ADALINE are linear classifiers that use a decision plane to separate two classes. The Perceptron employs a hard threshold activation function and updates its weights only when a misclassification occurs, aiming to find any separating hyperplane for (b) Perceptron and ADLINE both use a decision plane to separate two classes. Compare and contrast the two training techniques.

linearly separable data. As a result, when the problem is not linearly separable, the perceptron algorithm will fail to converge because it relies solely on classification errors. In contrast, ADALINE uses a linear activation function and applies the delta rule to minimise the mean squared error using gradient descent, allowing it to find a best-fit approximation to the target and resulting in a more stable training process even when the data are not perfectly linearly separable.

Question 3: [30 marks]

In a task to analyze the water quality of lakes, it is known that lakes come from one of two classes, natural lake C_1 or artificial lake C_2 . Each water quality sample of a lake Q has two features, $Q = (q_1, q_2)$. An experiment has collected 6 samples and their feature values and classification are shown in the following table. Answer the following questions by showing details of the working process. An answer without the working process will not receive any marks.

$Q = (q_1, q_2)$		Class
q_1	q_2	C
2	2	C_1
7	7	C_2
2	4	C_1

4	3	C ₁
5	4	C ₂
6	5	C ₂
5	2.5	=?

- (a) Based on the data with known class, design a KNN classifier with K=3 by the Euclidian distance to classify the unknown sample Q=(5, 2.5) in the last row of the table.

[10 marks]

- (b) Suppose the data in the above table are collected in a way that the class labels are unknown, i.e., we do not know C in the last column. Cluster the data into two clusters using K-Means algorithm with Euclidian distance and set the initial cluster centres as $W_1=(2, 2)$ and $W_2=(4, 3)$.

1.Business Understanding

[20 mark]

2.Data Understanding

3.Data Preparation

Question 4:**[28 marks]**

4.Modeling

- (a) Please list the 6 steps of Cross Industry Standard Process of Data Mining (CRISP-DM).

5.Evaluation

6.Deployment

[6 marks]

- (b) Both classification and regression are machine learning techniques for prediction from collected data. Compare and contrast the two techniques.

Both classification and regression are supervised learning techniques used for prediction from data. Classification predicts discrete class labels and aims to assign inputs to predefined categories, whereas regression predicts continuous numerical values by fitting a function to the data. As a result, classification focuses on decision boundaries and classification accuracy, while regression focuses on minimising numerical prediction error such as mean squared error.

[4 marks]

- (c) In Bayesian learning, one often adopt m-estimate of probability, whose formula is given below.

$$\frac{N_c + mP}{N + mP}$$

What N_c , N , m , P stands for? How to choose P ? What is the advantage of adopting m-estimate?

In m-estimate probability estimation, N_c denotes the number of training samples belonging to a given class, N is the total number of relevant training samples, m represents the equivalent sample size, which determines how heavily to weight P relative to the observed data, and P is the prior estimate of the probability we wish to estimate. The prior probability P is often chosen as a uniform distribution when no prior knowledge is available.

[8 marks]

- (d) We often use kernel functions together with SVM for better classification performance. Briefly describe what is a kernel function, what problem it tackles and why it can tackle the problem.

The advantage of using m-estimate is that it avoids poor estimation and zero probabilities and provides more robust probability estimates, especially when the training data are sparse.

[6 marks]

- (e) Recurrent Neural Network (RNN) is often used when predicting a sequence of outputs from a sequence of input data. Briefly describe the idea of RNN. You may draw a simple diagram to help illustrate the idea.

d) A kernel function implicitly maps input data from the original input space into a higher-dimensional feature space without explicitly computing the mapping. It is used to tackle the problem that data are not linearly separable in the original space. By transforming the data into a higher-dimensional space, the kernel function makes the data linearly separable, allowing SVM to construct a linear decision boundary in the feature space, which corresponds to a nonlinear boundary in the original space. A Recurrent Neural Network (RNN) is a neural network designed to model sequential data by maintaining a hidden state that captures information from previous time steps. At each time step, the network takes the current input and the previous hidden state to produce the current output and update the hidden state. By sharing weights across time steps, RNNs are able to learn temporal dependencies in sequence data and are therefore suitable for tasks involving sequences of inputs and outputs.

[4 marks]

End